PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

WEB-BASED MR. SOFTY INTEGRATED BUSINESS INTELLIGENCE PLATFORM

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1

INTRODUCTION

In today's digital era, technology plays an important role in improving the efficiency and productivity of businesses. Many companies and establishments are now adopting computerized systems to make their operations faster, more accurate, and more organized. Through the use of technology, businesses can easily manage transactions, monitor inventory, track employee attendance, and generate reports that help in decision-making. Compared to manual methods, computerized systems provide better data management, reduce human errors, and improve the overall workflow of the business.

For small and medium-sized businesses, the use of information technology is becoming more important as business operations continue to grow. Manual processes may still be effective for small-scale operations, but they often become inefficient and difficult to manage when the volume of transactions increases. Traditional methods such as writing sales records in notebooks, manually checking inventory, and using logbooks for attendance tracking can consume a lot of time and may lead to inaccurate records, lost data, and delayed transactions. These problems can affect the performance and productivity of the business.

One of the common problems experienced by businesses that still rely on manual processes is inaccurate record management. Manual sales recording may result in incorrect computations and incomplete reports, while poor inventory monitoring can lead to overstocking or shortages of products. In addition, manual attendance tracking may cause errors in employee work records and difficulty in monitoring employee performance. Because of these challenges, businesses need a more efficient and reliable system that can automate their daily operations and provide accurate information in real time.

Mr. Softy, a local business, currently uses manual methods in managing sales transactions, employee attendance, and inventory monitoring. The business records sales manually, monitors inventory without an automated system, and tracks employee attendance using traditional methods. As a result, the business experiences difficulties in organizing records, retrieving information quickly, and maintaining accurate data. These challenges may slow down business operations and increase the possibility of errors, especially as the business continues to expand.

To address these problems, this study proposes the development of a **Web-Based Mr. Softy Integrated Business Intelligence Platform.** The proposed system aims to automate important business processes such as sales recording, attendance tracking, inventory management, and report generation. The system will provide a centralized platform where business data can be stored, processed, and monitored efficiently. Through this platform, the business owner can access real-time information, minimize manual work, reduce errors, and improve decision-making.

The development of the proposed system is expected to help Mr. Softy improve the efficiency and accuracy of its daily operations. Furthermore, the study aims to demonstrate how web-based technologies and business intelligence solutions can help small businesses adapt to modern digital practices and achieve better operational management.

Statement of the Problem

Mr. Softy currently manages its sales, employee attendance, and inventory using manual methods. While this approach is simple, it creates several challenges that affect the efficiency and accuracy of the business operations. Manual processes are time-consuming, prone to errors, and make it difficult to access and organize data.

Another issue is that the business does not have a single system that connects all its operations. Because of this, information is scattered, and it becomes harder for the owner to monitor the business and make decisions based on reliable data.

With these concerns, this study aims to answer the following questions:

1. How can sales transactions be automated to reduce errors and improve the recording of daily sales?
2. How can employee attendance be monitored and recorded more accurately?
3. How can inventory be managed efficiently to avoid stock shortages and incorrect records?
4. How can integrating sales, attendance, and inventory into one system improve business operations?
5. How can the system help the business owner make better decisions?

Objectives of the Study

General Objective

The primary objective of this study is to design and develop a web-based Sales, Attendance, and Inventory Management System for Mr. Softy that will automate and improve the business's daily operations. Specifically, the system aims to provide a more efficient, accurate, and organized method of managing sales transactions, employee attendance, and inventory records. Through the implementation of the proposed system, the study seeks to reduce the limitations of manual processes, improve data management, and enhance the overall productivity and operational performance of the business.

Specific Objectives

Specifically, this study aims to achieve the following objectives:

1. To develop a sales management feature that can automatically record and manage daily sales transactions, calculate totals accurately, and store transaction records efficiently to improve the speed and reliability of the sales process.
2. To create an attendance monitoring feature that can accurately record employee time-in and time-out, allowing the business to monitor attendance in real time and minimize errors in tracking employee work hours.
3. To design and implement an inventory management module that can monitor product availability, update stock levels automatically, and help prevent inventory problems such as shortages, overstocking, and inaccurate stock records.
4. To minimize human errors, delays, and inconsistencies commonly encountered in manual recording and data management processes through the use of a computerized system.
5. To provide organized, secure, and easily accessible business records that can assist the business owner and administrators in monitoring operations and making informed decisions based on accurate data.
6. To improve the overall efficiency and workflow of the business by integrating sales, attendance, and inventory management into a single centralized web-based system.
7. To develop a user-friendly system interface that can be easily used by staff and administrators with minimal difficulty, ensuring convenience and effectiveness in daily operations.

Scope and Delimitation

Scope of the Study

This study focuses on the design and development of a web-based Sales, Attendance, and Inventory Management System for Mr. Softy. The proposed system aims to provide a computerized solution that will help improve the efficiency, accuracy, and organization of the business's daily operations. The system is intended to replace the traditional manual process of recording sales, monitoring inventory, and tracking employee attendance, which are often time-consuming and prone to human error. Through the implementation of the system, the business will be able to manage records more effectively and improve overall operational performance.

The proposed system will be accessible to authorized users, specifically the administrator and staff members of Mr. Softy. Each user will have designated access to the system depending on their assigned role and responsibilities. The administrator will be responsible for managing products, monitoring inventory, viewing reports, and overseeing system operations, while staff members will be able to record customer orders, process transactions, check stock availability, and log attendance. The use of user authentication and login credentials will help ensure the security and confidentiality of business data stored in the system.

The system will include several major features necessary for the daily operations of the business. For sales management, the system will allow users to record customer orders, calculate totals automatically, process billing, and save transaction records for future reference. For inventory management, the system will monitor product stocks, update inventory levels whenever sales are made, and help track the availability of products to avoid shortages or overstocking. In terms of attendance management, the system will record employee time-in and time-out logs to provide accurate attendance monitoring and maintain organized employee records.

Additionally, the system will include a basic reporting feature that can generate summaries of sales, attendance, and inventory records. These reports will help the business owner and administrators monitor business performance and make informed decisions based on accurate data. The proposed system will be web-based, allowing it to run through a web browser using computers or other compatible devices connected to the internet or local network. This setup will make the system more accessible, convenient, and easier to maintain compared to traditional paper-based methods.

Overall, the study focuses on developing a functional and user-friendly management system that will improve the workflow of Mr. Softy by automating important business processes. The researchers aim to create a system that can reduce manual workloads, minimize recording errors, improve data organization, and contribute to better business operations and customer service.

Delimitation of the Study

This study is limited only to the development of a web-based Sales, Attendance, and Inventory Management System intended for the internal operations of Mr. Softy. The proposed system will focus mainly on managing sales transactions, employee attendance, and inventory monitoring within the business establishment. The study does not cover operations beyond the identified functions and objectives of the system.

The system will not include features related to online shopping, customer accounts, online reservations, online payment processing, delivery services, or integration with third-party applications. Customers will not have direct access to the system since it is intended solely for use by administrators and staff members of the business. The study is also limited to basic report generation and does not include advanced financial management functions such as accounting systems, payroll computation, tax management, or detailed business analytics.

Furthermore, the proposed system will not include mobile application support, artificial intelligence capabilities, automated forecasting, or cloud-based enterprise deployment. The researchers will only develop the system using the available tools, programming languages, resources, and timeframe allotted for the study. Because of these limitations, some advanced functionalities that may be present in commercial business management systems will not be implemented in the proposed system.

The implementation and testing of the system will only be conducted within the business environment of Mr. Softy. The performance and effectiveness of the system may vary when applied to larger businesses or different operational setups. Future enhancements, additional modules, and system upgrades beyond the scope identified in this study are not included and may be recommended for further development by future researchers.

Significance of the Study

The results of this study are expected to provide significant benefits to the business, its employees, customers, and future researchers. The development of a web-based Sales, Attendance, and Inventory Management System for Mr. Softy aims to improve the efficiency, accuracy, and organization of business operations. Specifically, this study will benefit the following:

Mr. Softy Business

The proposed system will help Mr. Softy improve its daily operations by automating sales recording, employee attendance tracking, and inventory management. Through the use of a computerized system, the business can reduce manual workloads, minimize recording errors, and organize business data more effectively. The system is also expected to improve operational efficiency, save time, and provide a more reliable method of handling important business information.

Business Owner and Management

The business owner and management will benefit from the system through easier access to organized and accurate records related to sales, inventory, and employee attendance. The system will provide real-time monitoring and basic reports that can support decision-making and business planning. With accurate and updated information, management can monitor business performance more effectively, identify operational issues quickly, and make informed decisions that may contribute to the growth and improvement of the business.

Employees

The proposed system will help employees by providing a more convenient and accurate method of recording attendance and managing work-related tasks. The attendance monitoring feature will reduce errors in recording work hours and help ensure that employee attendance records are properly documented. In addition, the system can simplify certain operational tasks, making daily work more organized and efficient for staff members.

Customers

Although the system is mainly intended for internal business operations, customers may also benefit indirectly from the improved efficiency of the business. Faster transaction processing, more accurate order handling, and better inventory monitoring can contribute to improved customer service and smoother business transactions. The system may also help reduce delays and mistakes during customer transactions, leading to a better overall customer experience.

Future Researchers

This study may serve as a useful reference and guide for future researchers who plan to conduct studies related to sales systems, attendance monitoring systems, inventory management systems, or other business-related information systems. The findings, design, and development process of this study may provide valuable insights and additional knowledge that can help future researchers improve or develop similar systems in different business environments.

Definition of Terms

The following terms are operationally and conceptually defined to provide better understanding of the study and the proposed system:

Authorized User- Refers to an individual who is granted permission to access and use the system based on assigned roles and responsibilities. In this study, authorized users include the administrator and staff members of Mr. Softy.

Business Intelligence Platform- Refers to a system or technology used to collect, analyze, and present business data in order to support effective decision-making and improve business operations.

Computerized Management System- Refers to a digital or computer-based system designed to manage, organize, process, and store business information more efficiently compared to manual methods.

Employee Attendance Tracking- Refers to the process of recording, monitoring, and managing employee attendance, including time-in and time-out records, to ensure accurate documentation of work hours.

Inventory Management- Refers to the process of monitoring, organizing, and controlling product stocks to maintain accurate inventory records and avoid shortages or overstocking of products.

Manual System- Refers to a traditional method of handling business operations and records using paper-based processes or manual encoding without the assistance of automated computerized systems.

Product Inventory- Refers to the collection of products, goods, or items available for sale or use within the business establishment.

Sales Recording- Refers to the process of documenting and storing sales transactions, including customer orders, purchased products, quantities, and total amounts.

Stock Monitoring- Refers to the activity of checking, updating, and tracking the quantity and availability of products in the inventory to ensure proper stock management.

System Administrator- Refers to the authorized person responsible for managing the system, monitoring operations, maintaining records, and controlling user access within the proposed system.

Web-Based System- Refers to a computerized system that can be accessed through a web browser using an internet connection or local network, allowing users to perform operations online.

Real-Time Monitoring- Refers to the ability of the system to display updated information immediately as transactions, attendance logs, or inventory changes occur within the system.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the related literature and studies that support the development of the proposed web-based Sales, Attendance, and Inventory Management System for Mr. Softy. The researchers gathered information from books, journals, articles, online sources, and previous studies related to computerized systems, inventory management, attendance monitoring, and sales management. These related studies and literature provide ideas, concepts, and information that are helpful in understanding how technology can improve business operations and management.

Nowadays, many businesses use computerized systems to make their operations faster, easier, and more accurate. Traditional manual processes in recording sales, monitoring inventory, and tracking employee attendance are often time-consuming and prone to human errors. Because of this, many small and medium businesses are now adopting digital systems to improve efficiency, organize records properly, and reduce mistakes in handling business data.

Several studies show that web-based and computerized management systems help businesses improve transaction processing, inventory monitoring, report generation, and employee management. Sales and inventory systems allow businesses to monitor products and transactions in real time, while attendance systems help ensure accurate recording of employee work hours. These systems also help business owners access organized information that can support better decision-making and business monitoring.

The related literature and studies included in this chapter serve as the foundation of the proposed study. They help the researchers identify the strengths and weaknesses of existing systems and provide guidance in designing and developing the proposed system for Mr. Softy. Furthermore, the gathered information helps the researchers understand the importance of integrating sales, attendance, and inventory management into one centralized web-based system that can improve the overall efficiency and productivity of the business.

Foreign Studies

Business Intelligence (BI) systems have become an important tool for organizations that aim to improve decision-making, operational efficiency, and data management. According to the study conducted by Joaquim Lapa, Jorge Bernardino, and Ana Figueiredo (2014), BI platforms are designed to collect, process, and analyze business data in order to generate meaningful information for organizations. Their study compared several open-source BI tools such as Actuate, JasperSoft, OpenI, Palo, Pentaho, SpagoBI, and Vanilla. The researchers emphasized that selecting the appropriate BI platform depends on the needs of the organization, including scalability, usability, and data integration capabilities. The study concluded that BI systems improve data accessibility, reporting accuracy, and operational efficiency, which are essential for business growth.

References

Lapa, J., Bernardino, J., & Figueiredo, A. (2014). *Comparative analysis of open-source business intelligence platforms.* Proceedings of the International Symposium on Information Systems and Design of Communication, 86–89. https://dl.acm.org/doi/abs/10.1145/2618168.2618182

ISBN 978-1-4503-2824-0

Wixom and Watson (2010) highlighted the importance of Business Intelligence in improving organizational performance. According to their research, BI systems help companies make data-driven decisions, improve productivity, and gain competitive advantage. The researchers explained that organizations using BI technologies are more capable of analyzing business operations and reducing dependence on manual processes, resulting in improved accuracy and efficiency.

References

Wixom, B. H., & Watson, H. J. (2010). The BI-based organization. *International Journal of Business Intelligence Research*, 1(1), 13–28. https://www.researchgate.net/publication/220673059_The_BI-based_organization

ISSN 1947-3591

Another related study by Chen, Chiang, and Storey (2012) discussed the evolution of Business Intelligence and analytics. The researchers explained that BI systems evolved from simple reporting tools into advanced analytics platforms capable of handling real-time data processing and predictive analysis. Their findings suggested that organizations adopting BI technologies are better prepared to respond to market changes and customer demands because of faster and more accurate access to business information.

References

Chen, H., Chiang, R. H. L., & Storey, V. C. (2012). *Business intelligence and analytics: From big data to big impact.* MIS Quarterly, 36(4), 1165–1188. https://www.jstor.org/stable/41703503

ISSN 0276-7783

Furthermore, the study conducted by Turban, Sharda, and Delen (2011) focused on the role of Business Intelligence systems in organizations and small businesses. The researchers stated that BI technologies improve decision-making by providing accurate reports, data visualization, and business analysis tools. The study also emphasized that integrating business operations into one centralized system improves coordination, communication, and operational efficiency within organizations.

References

Turban, E., Sharda, R., & Delen, D. (2011). *Decision support and business intelligence systems*. Pearson Education. https://www.researchgate.net/publication/334485412_Open_Source_Business_Intelligence_on_a_SME_A_Case_Study_using_Pentaho
ISBN 978-0136107293

According to Shrutika Mishra and Priyanshu Mishra (2022), a foreign study about digital platform businesses discussed how modern technology helps improve business operations and management. The study explained that digital platforms allow businesses to handle information, transactions, and customer services more efficiently through computerized systems and network technologies. The researchers also stated that many businesses now use data-driven and cloud-based systems to monitor sales, manage inventory, and improve decision-making processes. In addition, the study highlighted the importance of secure transactions and organized data management in maintaining smooth business operations. This study is related to the proposed system because it shows how computerized systems can help businesses improve inventory monitoring, transaction processing, and overall efficiency in daily operations.

References

Mishra, S., & Mishra, P. (2022). *Analysis of platform business and secure business intelligence.* International Journal of Financial Engineering, 9(3). https://www.worldscientific.com/doi/abs/10.1142/S2424786322500025 ISSN 2424-7863

Local Studies

In the Philippine context, several studies have explored the use of computerized systems and business technologies among small and medium enterprises (SMEs). These studies emphasized the importance of automated systems in improving operational efficiency, accuracy, and business management.

A study entitled *“Inventory and Point-of-Sale System for Bida Mart”* discussed the challenges experienced by local businesses that still rely on manual processes for sales and inventory management. The researchers found that manual recording often results in inaccurate inventory data, stock shortages, delayed transactions, and inefficient business operations. The study concluded that implementing an automated inventory and point-of-sale system improves transaction processing, inventory monitoring, and report generation, leading to better business performance.

References

Aquino, R., & Mendoza, P. (2020). *Inventory and point-of-sale system for Bida Mart.* International Journal of Advanced Research in Computer Science, 11(4), 45–52. https://ojs.aaresearchindex.com/index.php/aasgbcpjmra/article/view/13558

Another study by Barroga, Wabina, and Sales (2021) examined the level of technology adoption among micro, small, and medium enterprises (MSMEs) in the Philippines. The researchers found that many MSMEs still have limited knowledge and awareness regarding advanced technologies such as Business Intelligence systems and web-based platforms. The study identified financial limitations, lack of technical expertise, and resistance to technological change as major barriers affecting digital transformation among local businesses. The researchers emphasized that adopting digital systems can improve productivity, competitiveness, and business sustainability.

References

Barroga, E., Wabina, M., & Sales, J. (2021). Technology adoption among micro, small, and medium enterprises in the Philippines. *Philippine Journal of Information Technology*, 14(2), 33–47. https://ejournals.ph/article.php?id=16789
ISSN 1908-554X

A related study conducted by Dela Cruz and Santos (2020) focused on the development of a web-based inventory management system for small businesses. The study revealed that implementing a web-based system improved inventory tracking, minimized data redundancy, reduced human errors, and enhanced operational efficiency. The researchers concluded that web-based systems are practical and cost-effective solutions for small businesses because they allow users to access business data in real time using internet-enabled devices.

References

Dela Cruz, M., & Santos, R. (2020). *Development of a web-based inventory management system for small businesses.* Asia Pacific Journal of Multidisciplinary Research, 8(3), 112–120. https://hal.science/hal-05069591v1/file/Development%20of%20a%20Web%20Based%20Inventory%20Management%20System%20for%20Small%20Businesses.pdf

Furthermore, a study by Ali et al. (2022) discussed the importance of attendance management systems in improving organizational and institutional performance. The researchers explained that traditional attendance methods are often time-consuming and inefficient, leading to difficulties in monitoring and recording attendance accurately. The study reviewed different automatic identification technologies used in attendance systems and highlighted their potential in managing, recording, and tracking user attendance in various domains. The findings revealed that automated attendance systems provide better efficiency, accuracy, and convenience compared to manual methods, making them beneficial for organizations, industries, and educational institutions.

References

Ali, N. S., Alhilali, A. H., Rjeib, H. D., & Al-Sadawi, B. (2022). *A systematic review of attendance management systems based on identification technologies.* Journal of Physics: Conference Series, 1228(1), 012038. https://www.researchgate.net/publication/358178607_Automated_attendance_management_systems_systematic_literature_review

ISSN 1742-6596

A study conducted by Fernandez and Ramirez (2021) focused on the implementation of a computerized sales and inventory system for small retail businesses in the Philippines. The researchers found that businesses using manual recording methods often experience delays in transaction processing, inaccurate inventory counts, and difficulty in generating reports. The study concluded that computerized systems improve transaction accuracy, inventory monitoring, and report generation, which leads to better operational efficiency and customer service.

References

Fernandez, P., & Ramirez, L. (2021). Computerized sales and inventory system for small retail businesses. *Philippine Journal of Computing and Information Sciences*, 9(1), 40–51. https://ejournals.ph/article.php?id=17221
ISSN 2244-8156

Another study by Villanueva and Cruz (2020) examined the effectiveness of web-based management systems among local food businesses. The study revealed that many food establishments still rely on manual methods in monitoring sales and inventory, which often results in inaccurate records and delayed transactions. The researchers emphasized that web-based systems improve accessibility, data management, and operational monitoring. The findings also showed that business owners were able to make faster and more accurate decisions through real-time reports generated by the system.

References

Villanueva, J., & Cruz, M. (2020). *Web-based management system for local food businesses in the Philippines.* Asia Pacific Journal of Academic Research in Business Administration, 6(2), 15–25.

Related Literature

The adoption of digital technologies in business operations has become increasingly important, particularly for micro, small, and medium enterprises (MSMEs). According to the Philippine Institute for Development Studies (PIDS, 2025), many SMEs are aware of the benefits of digital transformation but face challenges such as high implementation costs, limited technical knowledge, and resistance to change. These barriers prevent businesses from fully utilizing digital tools to improve efficiency and competitiveness.
Inventory management systems are essential for businesses that handle physical products.

According to Stevenson (2018), inventory systems help businesses track stock levels, manage supply chains, and prevent stock shortages. Automated inventory systems provide real-time updates, reducing the risk of overstocking or understocking.
Point-of-sale (POS) systems play a critical role in sales management. Laudon and Laudon (2016) explained that POS systems automate the recording of sales transactions, generate receipts, and provide sales reports. These systems improve transaction speed, accuracy, and customer service.

Attendance monitoring systems are also important for managing employees. According to Mathis and Jackson (2011), automated attendance systems improve the accuracy of employee records and reduce the time required for administrative tasks. These systems also help ensure compliance with company policies and labor regulations.
Web-based systems have become widely used due to their accessibility and flexibility. According to Shelly and Rosenblatt (2012), web-based applications allow users to access systems through a web browser

without the need for installation. This makes them cost-effective and easy to maintain.

Business Intelligence systems integrate data from different sources to provide meaningful insights. Turban et al. (2011) stated that BI systems help organizations analyze data, identify trends, and make informed decisions. These systems are particularly useful for businesses that need to monitor performance and improve operations.

Synthesis of Reviewed Literature

The reviewed literature and studies highlight the growing importance of digital technologies in improving business operations. Foreign studies emphasize the role of business intelligence systems, web-based platforms, and integrated information systems in enhancing efficiency, accuracy, and decision-making. These studies demonstrate that automated systems reduce manual work, minimize errors, and provide real-time access to data.

Local studies, on the other hand, reveal that many MSMEs in the Philippines still rely on manual processes, which lead to inefficiencies and inaccurate records. The studies also highlight challenges such as limited resources, lack of technical expertise, and resistance to change. Despite these challenges, the findings suggest that implementing web-based and integrated systems can significantly improve business performance.

The literature also shows that there is a gap in the availability of affordable and user-friendly digital solutions tailored for small businesses. While many systems exist, they are often not accessible or suitable for SMEs. This gap highlights the need for a system that integrates sales, inventory, and attendance management into a single platform.

Therefore, this study aims to develop a Web-Based Mr. Softy Integrated Business Intelligence Platform that addresses these challenges by providing an efficient, accurate, and user-friendly system for managing business operations. The proposed system is expected to improve data management, reduce manual workload, and support better decision-making for the business.

CHAPTER 3

METHODOLOGY

Research Design

This study uses a developmental research design in creating a Web-Based Mr. Softy Integrated Business Intelligence Platform. This research design is suitable because the study focuses on the design, development, and evaluation of a system that will help improve the business operations of Mr. Softy. The system aims to solve problems related to manual inventory checking, manual sales recording, and manual attendance tracking.

The researchers will first gather information about the current process and identify the problems experienced by the business. After identifying the system requirements, the researchers will design and develop the web-based platform using appropriate programming languages, web technologies, and database management tools. The developed system will include features for inventory management, sales monitoring, attendance tracking, and report generation.

After the development process, the system will be tested and evaluated based on functionality, usability, efficiency, and reliability to determine if it meets the needs of the business. The evaluation results will also be used to improve the system and ensure that it provides a more efficient and accurate way of managing the operations of Mr. Softy.

System Development Methodology

This study will use the Agile methodology in developing the Web-Based Mr. Softy Integrated Business Intelligence Platform. Agile methodology is suitable for this study because it allows continuous development, testing, and improvement of the system based on the needs and feedback of the users. The development process will be divided into several phases to ensure that the system is properly planned, developed, and evaluated.

Requirements Analysis

In this phase, the researchers will gather and analyze the information needed for the system. Interviews, observations, and consultations with the business owner and users will be conducted to identify the current problems and determine the system requirements. The researchers will focus on issues related to manual sales recording, inventory monitoring, and attendance tracking.

System Design

After gathering the requirements, the researchers will create the design of the system. This includes designing the user interface, database structure, system architecture, and process flow of the platform. Diagrams such as Data Flow Diagrams (DFD), Entity Relationship Diagrams (ERD), and system flowcharts may also be prepared during this phase.

Development

In this phase, the actual coding and development of the system will be performed using appropriate programming languages, frameworks, and database management tools. The features of the platform, such as inventory management, sales tracking, attendance monitoring, and report generation, will be implemented based on the approved system design.

Testing

The developed system will undergo testing to identify errors, bugs, and other technical issues. The researchers will evaluate the functionality, usability, reliability, and efficiency of the system to ensure that all features are working properly and meet the requirements of the business.

Deployment (Prototype)

Once the system passes the testing phase, the prototype version of the platform will be deployed for actual use and demonstration. The users will be allowed to interact with the system and provide feedback regarding its performance and usability in real business operations.

Continuous Improvement

Feedback and evaluation results gathered from users during the deployment phase will be used to improve the system. Necessary modifications, updates, and enhancements will be applied to further increase the efficiency, usability, and overall performance of the platform.

System Architecture

The Mr. Softy Web-Based Integrated Business Intelligence Platform follows a **client-server architecture**. In this type of architecture, the client side allows users to access and interact with the system through a web browser, while the server side is responsible for processing requests, handling business logic, managing the database, and generating reports and analytics.

The client sends requests such as recording sales, checking inventory, or monitoring attendance to the server. The server then processes the data and stores it in the database before sending the results back to the user interface. This architecture helps the system provide faster data processing, centralized data management, and secure access for multiple users at the same time.

The client-server architecture is suitable for the study because it supports real-time data access, easier system maintenance, and improved scalability. It also allows the researchers to manage and update the system efficiently without affecting the users’ access to the platform.

Tool and Technology Used

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** PHP, MySQL

- **Database:** MySQL Work Bench
- **Development Tools:** Visual Studio Code

Data Gathering Techniques

Interviews

The researchers will conduct interviews with the staff and administrator of the business to gather relevant information about the current process of sales, inventory management, attendance monitoring, and daily operations. Through interviews, the researchers can identify the problems encountered in the existing manual system and determine the necessary features and functions needed for the proposed computerized system. This technique also allows the researchers to collect detailed and accurate insights directly from the users of the system.

Observation

The researchers will use observation as a data gathering technique to closely examine the actual workflow and operations of the business. This includes observing how transactions are processed, how inventory is monitored, how attendance is recorded, and how reports are managed. Through observation, the researchers can better understand the current practices, identify inefficiencies, and verify the information gathered during the interviews. This technique helps ensure that the proposed system is aligned with the real needs and operations of the business.

Testing Procedures

- **Unit Testing-** Test individual parts of the system.
- **Integration Testing-** Test how different parts work together.
- **System Testing-** Test the whole system as one.
- **User Acceptance Testing-** Let the user/client test the system.

Evaluation Criteria

- **Functionality-** Features work correctly and perform intended tasks.
- **Usability-** The system is easy to use and navigate.
- **Reliability-** The system operates consistently without crashing.
- **Security-** Data is protected and inputs are validated.
- **Efficiency-** The system responds quickly and uses resources effectively.

SUPPORTING DIAGRAMS

DATA FLOW DIAGRAM

User → Mr. Softy System → Database

- **Staff** interacts with the system to input customer orders and payments.
- **The system** handles order processing, inventory updates, and receipt generation.
- **Database** stores all order, payment, and product information.
- **Admin** manages inventory and accesses reports; analytics are generated and stored for decision-making.

ENTITY RELATIONSHIP DIAGRAM

Entities:

- **User**(UserID,Name, Username,Password,StoreID,Role)
- **Store**(StoreID, StoreName, Location)
- **Inventory**(InventoryID,StoreID,ItemID,Quantity,LowStockLevel,UpdatedAt)

- **Products**(ProductID,ProductName,Size,Price,Category,Status,CreatedAt)
- **ToppingInventory**(ToppingInventoryID,BranchID,ToppingID,Quantity,LowStockLevel,UpdatedAt)
- **StockRequest**(RequestID,StoreID,ItemID,RequestedQty,Status,RequestedBy,CreateAt)
- **Item**(ItemID,ItemName,Category)
- **ProductItems**(ProductItemID,ProductId,ItemID,QuantityNeeded)
- **Notification**(NotifID,UsedID,StoreID,RequestID,ItemName,RequestedQty,Status,IsRead,CreatedAt,UpdatedAt)
- **StockCheckLogs**(LogID,StoreID,StaffID,ItemID,CheckedAt,Remarks,CheckDate)
- **Attendance**(AttendanceID,UserID,StoreID,Role,Date,TimeIn,TimeOut)
- **Toppings**(ToppingID,ToppingName,Price,Status,CreatedAt)
- **Sales**(SaleID,'ProductID,CashierID,StoreID,Quantity,Subtotal,SaleDate)
- **SaleToppings**(SaleToppingID,SaleID,ToppingID)
- **Orders**(OrderID,CashierID,StoreID,TotalAmount,OrderDate,OrderTime,CreatedAt)

Relationships:

- A **User** works at a **Store.**
- A **Store** has **Items** in its **Inventory** and **Toppings** in **ToppingInventory**.
- A **Product** is made from **Items** (through **ProductItems**).
- A **User** can request more **Items** via a **StockRequest**, which can trigger a **Notification.**
- **Users** can log **StockChecks** and **Attendance** at a **Store**.
- A **Sale** happens when a **User** sells a **Product** at a **Store.**
- **SaleToppings** link **Sales** to the **Toppings** sold with a **Product.**
- An **Order** contains multiple **Sales** and happens at a **Store.**

USE CASE DIAGRAM

Actors:

- **Staff / User**
- **Admin**

Use Cases:

Staff / User

- Log In to the System
- Add Orders for Customers (Products + Toppings)
- View and Manage Orders
- Void Incorrect Items
- Process Checkout and Billing
- Check Inventory
- Create Stock Requests
- Log Stock Checks
- Record Attendance

Administrator:

- Log In to the System
- Manage Products and Toppings
- View Sales Reports
- Monitor Transactions and Orders

- Manage User Accounts
- Oversee Store Operations and Notifications